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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Introduction to Network Protocols

Overview of DNS, DHCP, HTTP/HTTPS, FTP, and SNMP. Observe DNS and HTTP traffic using Wireshark.

Task: Write a report on the role of DNS and HTTP in web communication.

- **1.Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.**

  - **Aim**
    - To understand the role of different network protocols and use Wireshark to capture and identify DNS and HTTP/HTTPS-related packets while accessing the Spotify website.

  - **Introduction**
    - Network protocols are a set of rules that allow computers and other devices to communicate and exchange information over a network.
    - When a user opens a website, different network protocols work together to establish communication between the user's device and the web server.

- Some commonly used network protocols are DNS, DHCP, HTTP, HTTPS, FTP, and SNMP. Wireshark is a network protocol analyzer that helps capture and examine network packets to understand how these protocols work.

➢ **Overview of Network Protocols**
➢ **1. DNS (Domain Name System)**
➢ DNS converts human-readable domain names into IP addresses. It helps a computer locate the server associated with a website.
➢ **Example:** When a user enters www.spotify.com in a browser, DNS helps find the IP address associated with the domain name.

➢ **2. DHCP (Dynamic Host Configuration Protocol)**
➢ DHCP automatically assigns network configuration information to devices, such as IP addresses, subnet masks, default gateways, and DNS server addresses.
➢ **Example:** When a laptop connects to a Wi-Fi network, the router's DHCP service can automatically assign an IP address to it.

➢ **3. HTTP (Hypertext Transfer Protocol)**
➢ HTTP is an Application Layer protocol used to transfer web pages and other resources between a browser and a web server. Standard HTTP commonly uses TCP port 80.
➢ **Example:** A browser sends an HTTP request to a web server to access a webpage.

➢ **4. HTTPS (Hypertext Transfer Protocol Secure)**

- HTTPS is the secure version of HTTP. It uses TLS encryption to protect communication between a browser and a web server. HTTPS commonly uses TCP port 443. HTTP/3 uses QUIC over UDP port 443.
- **Example:** When a user visits https://www.spotify.com, HTTPS helps protect the communication between the browser and the website.

- **5. FTP (File Transfer Protocol)**
- FTP is a protocol used to transfer files between a client and a server over a network. Traditional FTP commonly uses TCP ports 20 and 21.
- **Example:** An organization may use FTP to transfer files between computers and a file server.

- **6. SNMP (Simple Network Management Protocol)**
- SNMP is used to monitor and manage network devices such as routers, switches, printers, and servers.
- **Example:** A network administrator can use SNMP-based monitoring software to check the status and performance of network devices.

- **Role of DNS and HTTP in Web Communication**
- DNS and HTTP/HTTPS play important roles when a user opens a website.
- **Role of DNS**
  1. The user enters a website address, such as www.spotify.com, into the browser.
  2. The browser or operating system checks whether the IP address is already available in its cache.
  3. If necessary, a DNS query is sent to a DNS resolver.

4. The DNS resolver obtains the required DNS information and returns an IP address or other relevant DNS records.
5. The browser uses the resolved address to communicate with the website's server.

➢ **Role of HTTP/HTTPS**

1. After resolving the domain name, the browser establishes communication with the website's server.
2. The browser sends an HTTP request or establishes secure HTTPS communication.
3. The server processes the request and sends the requested webpage resources.
4. The browser receives and processes the response.
5. The webpage is displayed to the user.

➢ **Tools and Requirements**

- A laptop or computer
- Internet connection
- Wireshark software
- A web browser
- Spotify website: www.spotify.com

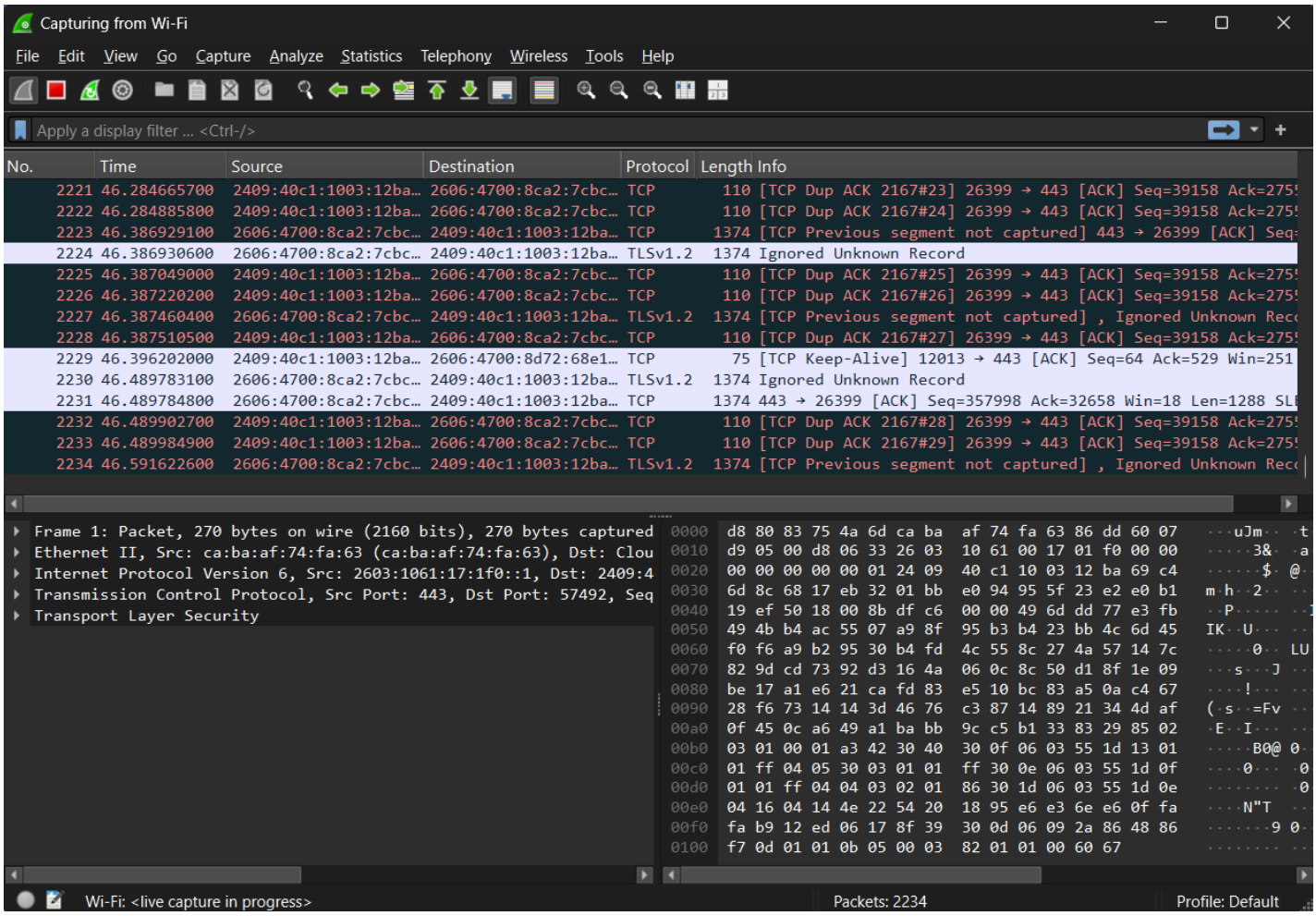
➢ **Procedure: Capturing Network Traffic Using Wireshark**

➢ **Step 1: Open Wireshark**

➢ Open Wireshark on your computer and identify the active network interface, such as Wi-Fi or Ethernet.

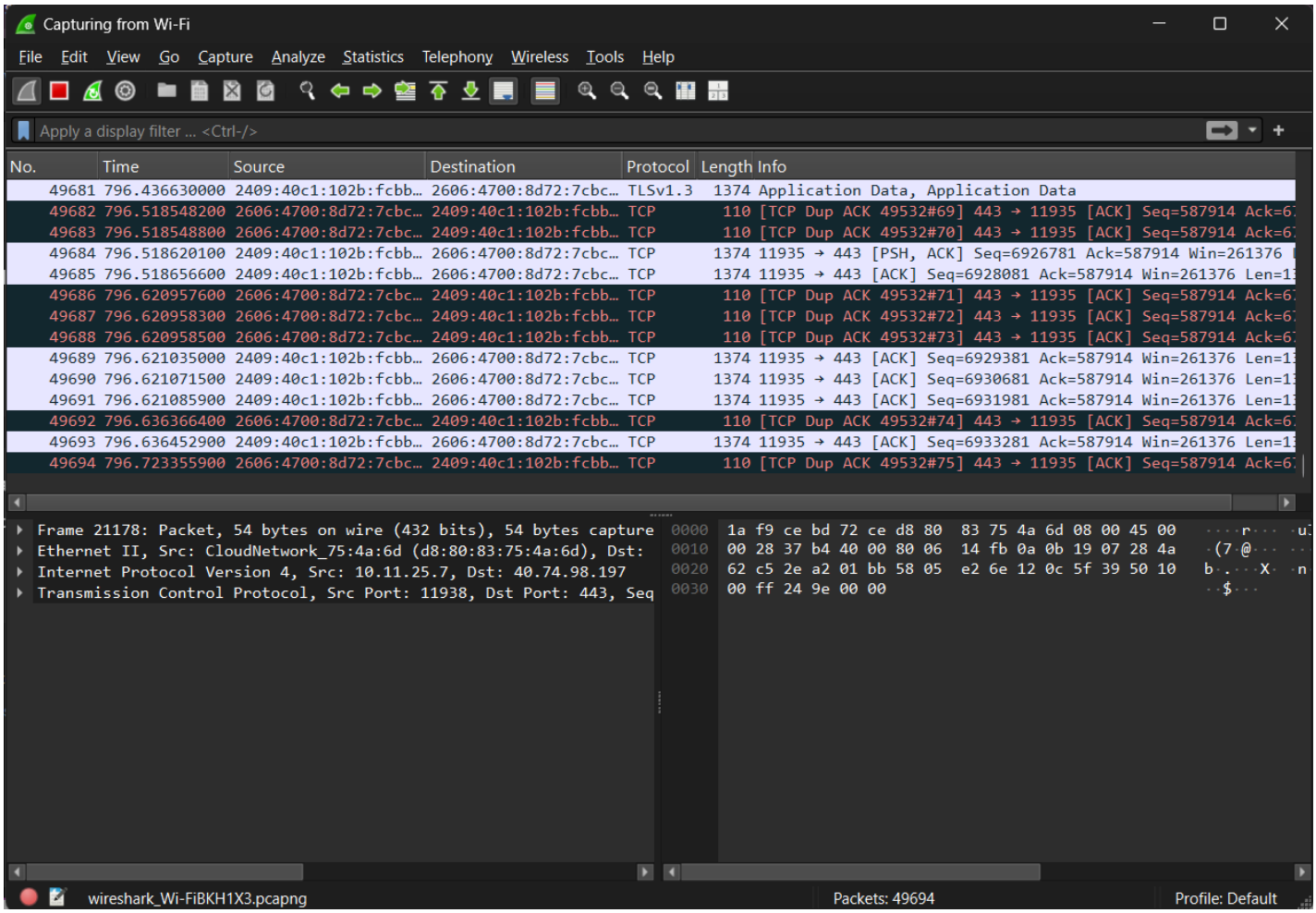
➢ **Step 2: Start Packet Capture**

- Double-click the active network interface to start capturing network traffic.



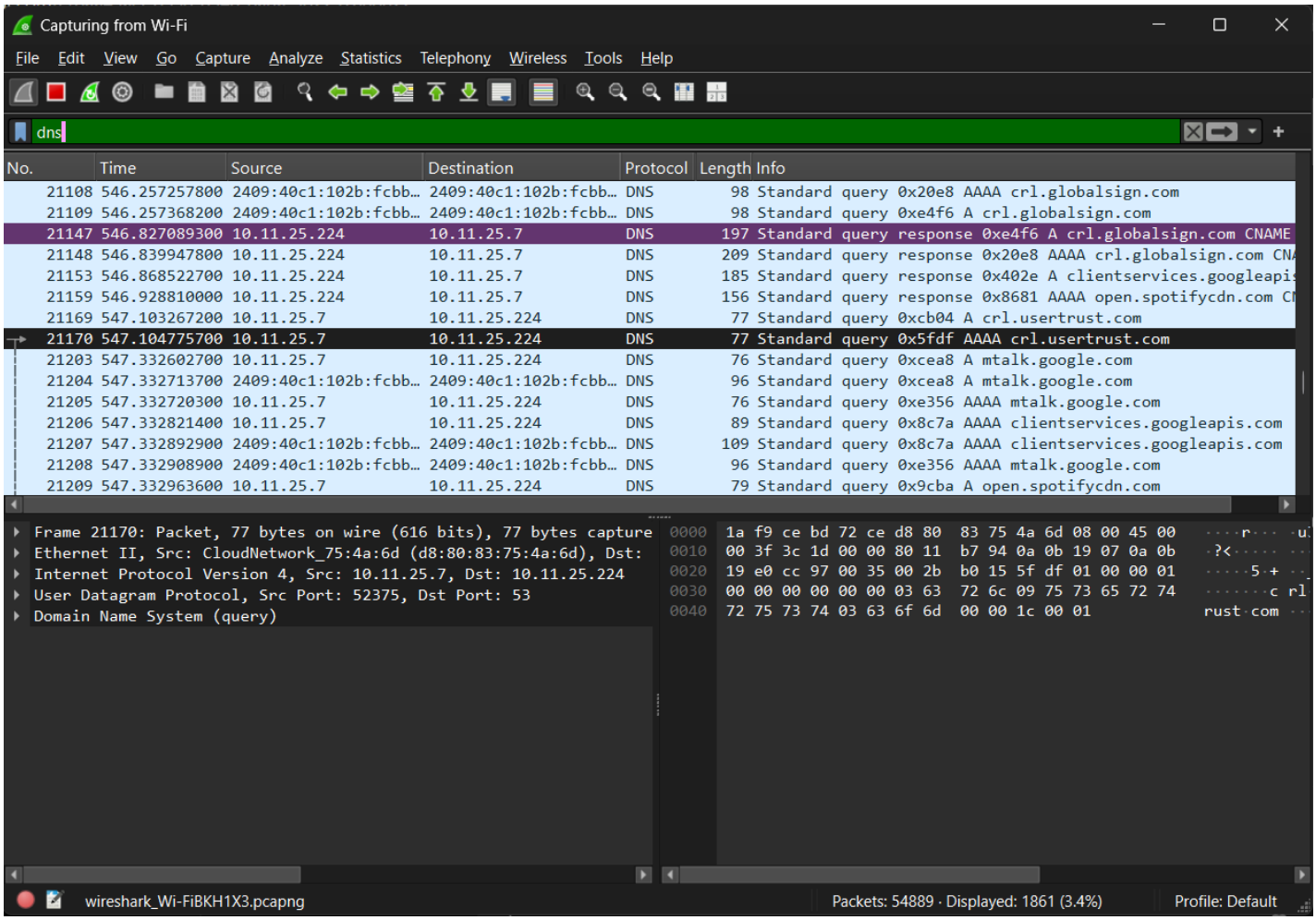
- **Step 3: Open Spotify Website**
- Open a web browser and enter:
- https://www.spotify.com
- Allow the website to load completely.
- **Step 4: Capture Network Traffic**

- Keep Wireshark running while the website loads. Continue capturing traffic for at least 30 seconds.



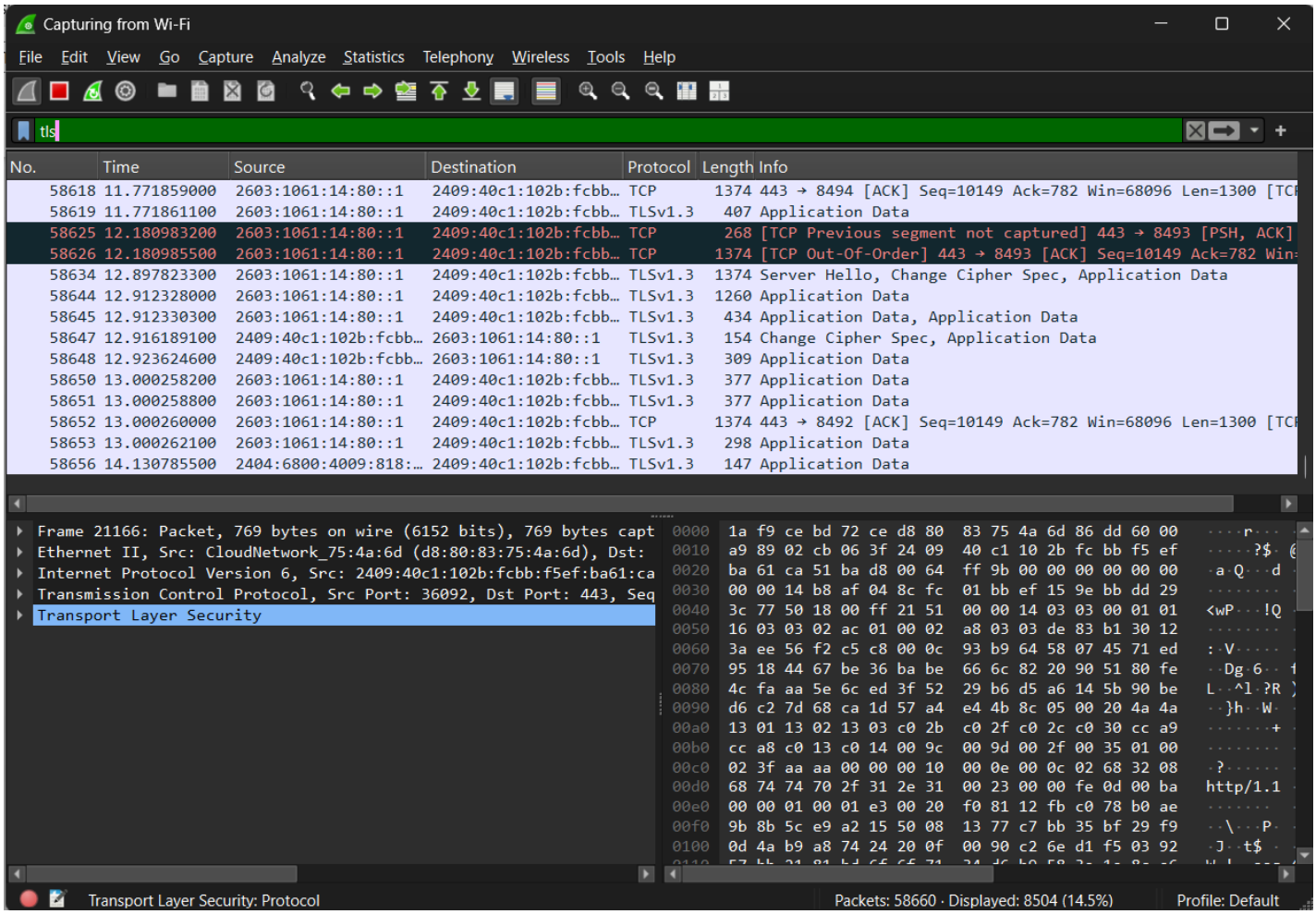
- **Step 5: Stop the Capture**
- Return to Wireshark and click the Stop button to stop packet capture.
- **Step 6: Identify DNS Packets**
- Enter the following display filter in Wireshark:
- dns
- This filter displays DNS packets, if any are present in the capture.

- Look for DNS queries and responses related to Spotify or its associated domains.



- **Step 7: Identify HTTP/HTTPS-Related Packets**
- Use the following filters:
  - http — displays packets identified as HTTP.
  - tls — displays TLS packets associated with secure communication.
  - quic — displays QUIC packets, which may be used by HTTP/3.

- Select individual packets and examine their details in the packet details pane.



- **Observation: DNS Packet Details**
- After applying the dns display filter, identify at least two DNS packets from your capture.

Sr. No.	Packet Type	Information to Record
1	DNS Query	Source IP, destination IP, queried domain name and packet number
2	DNS Response	Source IP, destination IP, response records or returned IP address and packet number

- **Explanation:** A DNS query is sent to request information about a domain name. The DNS response provides the requested DNS information, such as an IP address, when available.

- **Observation: HTTP/HTTPS Packet Details**
- After applying the appropriate display filter, identify at least two HTTP or HTTPS-related packets from your capture.

Sr. No.	Packet Type	Information to Record
1	HTTPS/TLS or QUIC Packet	Source IP, destination IP, protocol, packet number and available handshake information
2	HTTPS/TLS or QUIC Packet	Source IP, destination IP, protocol, packet number and available packet details

- **Explanation:** HTTPS-related traffic protects communication through encryption. Wireshark may display TLS handshake details and other packet metadata, but the encrypted webpage content is generally not readable.

- **Result**
- Network traffic was captured using Wireshark while accessing the Spotify website. DNS packets and HTTP/HTTPS-related packets were examined using suitable display filters. The packet details helped in understanding how DNS resolves domain names and how secure web communication takes place.

❖ **2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.**

- **Aim**
  - To perform a DNS lookup for www.flipkart.com using the nslookup command and identify the IP address returned by the DNS server.

- **Introduction**
  - **DNS (Domain Name System)** is a network service that converts human-readable domain names into IP addresses.
  - Computers use IP addresses to communicate with servers over a network. DNS makes it easier for users to access websites by allowing them to enter domain names instead of remembering numerical IP addresses.

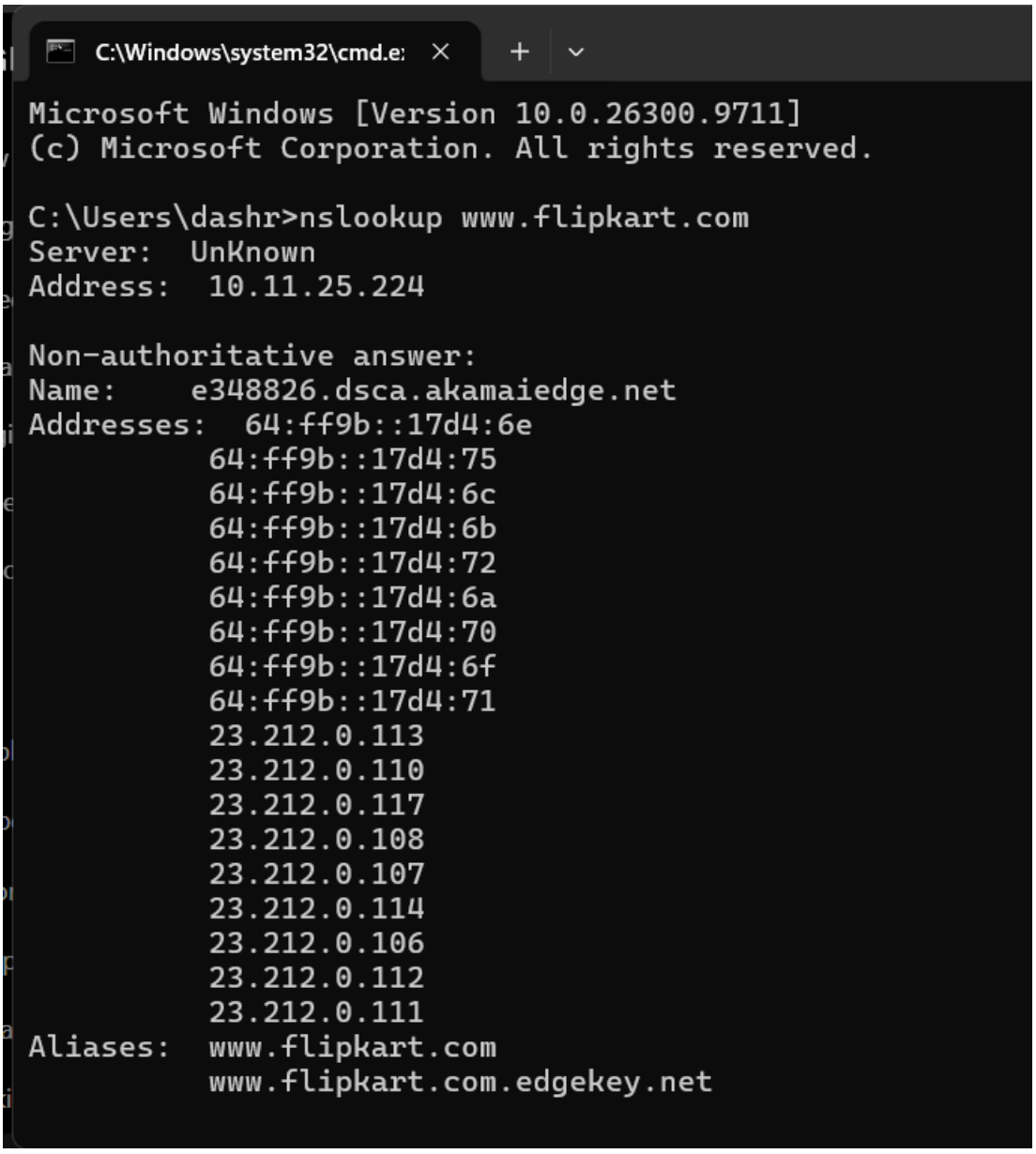
- The nslookup command is a network troubleshooting tool used to query DNS servers and obtain information about domain names.

➢ **Requirements**

- A computer or laptop
- Internet connection
- Windows Command Prompt or Terminal
- nslookup command

➢ **Procedure**

1. Open the Start menu in Windows.
2. Search for **Command Prompt** and open it.
3. Type the following command:
   nslookup www.flipkart.com
4. Press Enter.
5. The DNS server processes the query and returns the available DNS information.



- **Observation**
- The following command was used to perform the DNS lookup: **nslookup www.flipkart.com**
- **IP Address Returned:** 23.212.0.113

- **DNS Server:** Unknown

- **Explanation**
- When the nslookup command is executed, the computer sends a DNS query to a DNS server to find the IP address associated with www.flipkart.com.
- The DNS server searches for the requested domain information and returns the available IP address records.
- This means that the domain name has been resolved into an IP address that a computer can use to communicate with the website's server or associated network service.

- **Result**
- The DNS lookup for www.flipkart.com was performed successfully using the nslookup command. The IP address returned by the DNS server was identified and recorded.

❖ **3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.**

➢ **Aim**

- To understand the difference between HTTP and HTTPS by identifying three key differences and explaining a real-world example of a website that uses HTTPS for secure communication.

➢ **Introduction**

- **HTTP (Hypertext Transfer Protocol)** is an Application Layer protocol used for communication between a web browser and a web server. It allows users to access web pages and transfer information over a network.
- **HTTPS (Hypertext Transfer Protocol Secure)** is the secure version of HTTP. It uses TLS (Transport Layer Security) encryption to protect communication between a browser and a web server.

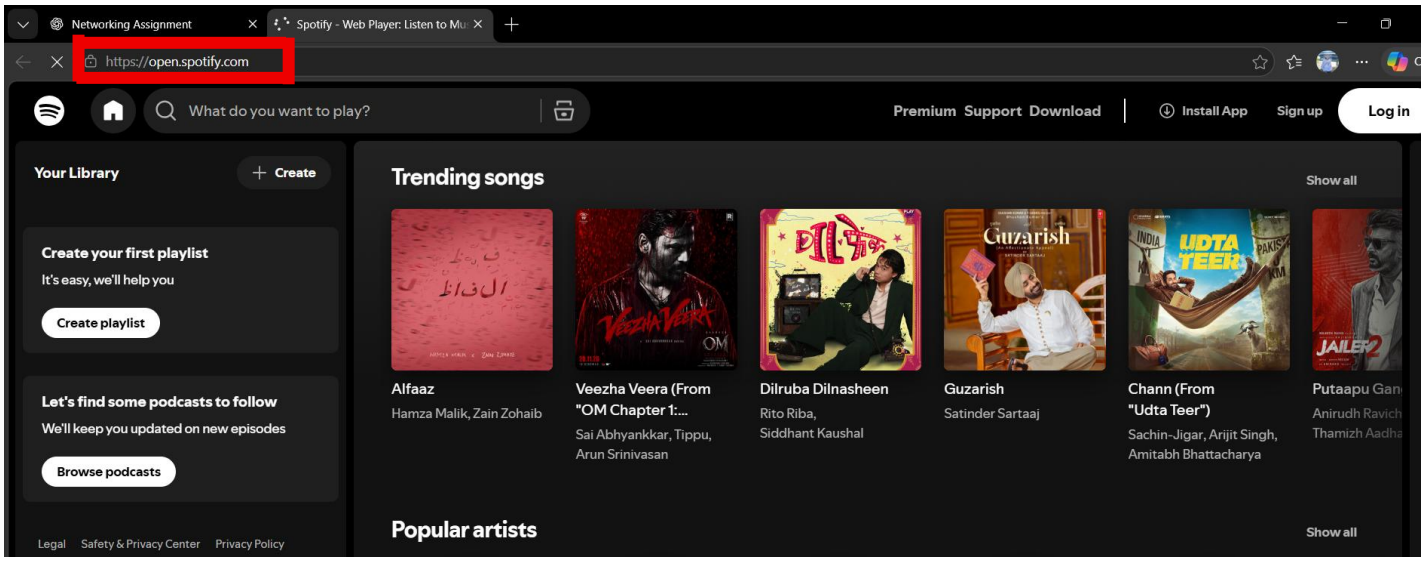
➢ Both protocols are used for web communication, but HTTPS provides additional security and protection for transmitted information.

➢ **Three Key Differences Between HTTP and HTTPS**

Sr. No.	**HTTP**	**HTTPS**
1. Security	HTTP transfers data without TLS encryption, so the communication is not protected by HTTPS security.	HTTPS uses TLS encryption to protect

		data exchanged between the browser and server.
2. Port Number	HTTP commonly uses TCP port 80.	HTTPS commonly uses TCP port 443.
3. Data Protection	Data transmitted through HTTP can be read or modified more easily if intercepted.	HTTPS provides encryption and helps protect data from unauthorized reading or modification during transmission.

- **Real-World Example of HTTPS**
- **Example: Spotify Website**
- The Spotify website (https://www.spotify.com) uses HTTPS to provide secure communication between the user's browser and its web servers.
- When a user visits Spotify, HTTPS helps protect the information exchanged between the browser and the website through TLS encryption.
- This helps protect user communication from unauthorized access and tampering during transmission.



- **Result**
- Three key differences between HTTP and HTTPS were identified based on security, port numbers, and data protection. Spotify was selected as a real-world example of a website that uses HTTPS for secure communication.

❖ **4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Look for packets with 'Standard query' and 'Standard query response' in the Info column.**

➢ **Aim**

- To capture and analyze DNS traffic using Wireshark while accessing a new website and understand the sequence of DNS queries and responses.

➢ **Introduction**

- **DNS (Domain Name System)** is a network protocol that converts human-readable domain names into IP addresses. It helps computers locate websites and other network services over the Internet.
- Wireshark is a network protocol analyzer that allows us to capture and examine DNS packets. By observing DNS queries and responses, we can understand how a browser obtains the IP address of a website.

➢ **Tools and Requirements**

- A computer or laptop
- Internet connection
- Wireshark software
- A web browser
- A website that has not been visited recently

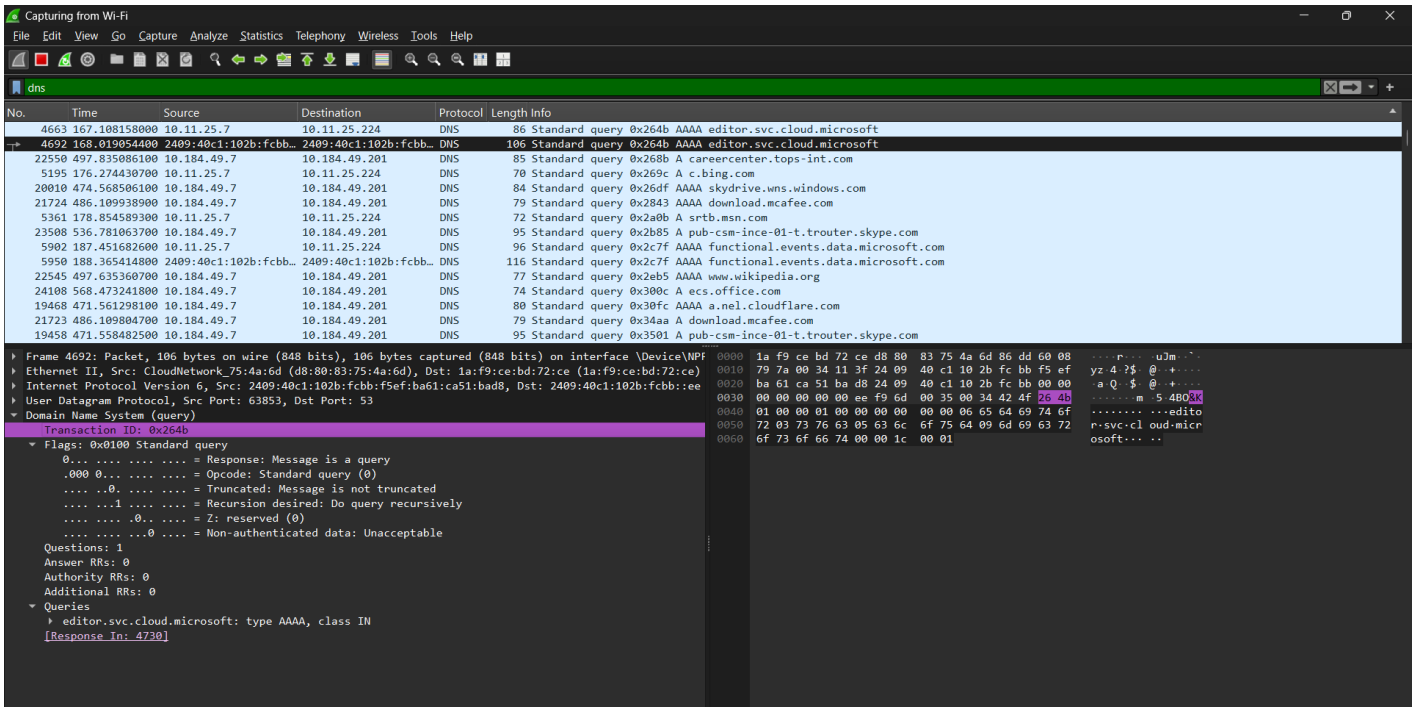
➢ **Procedure**

1. Open Wireshark on the computer.

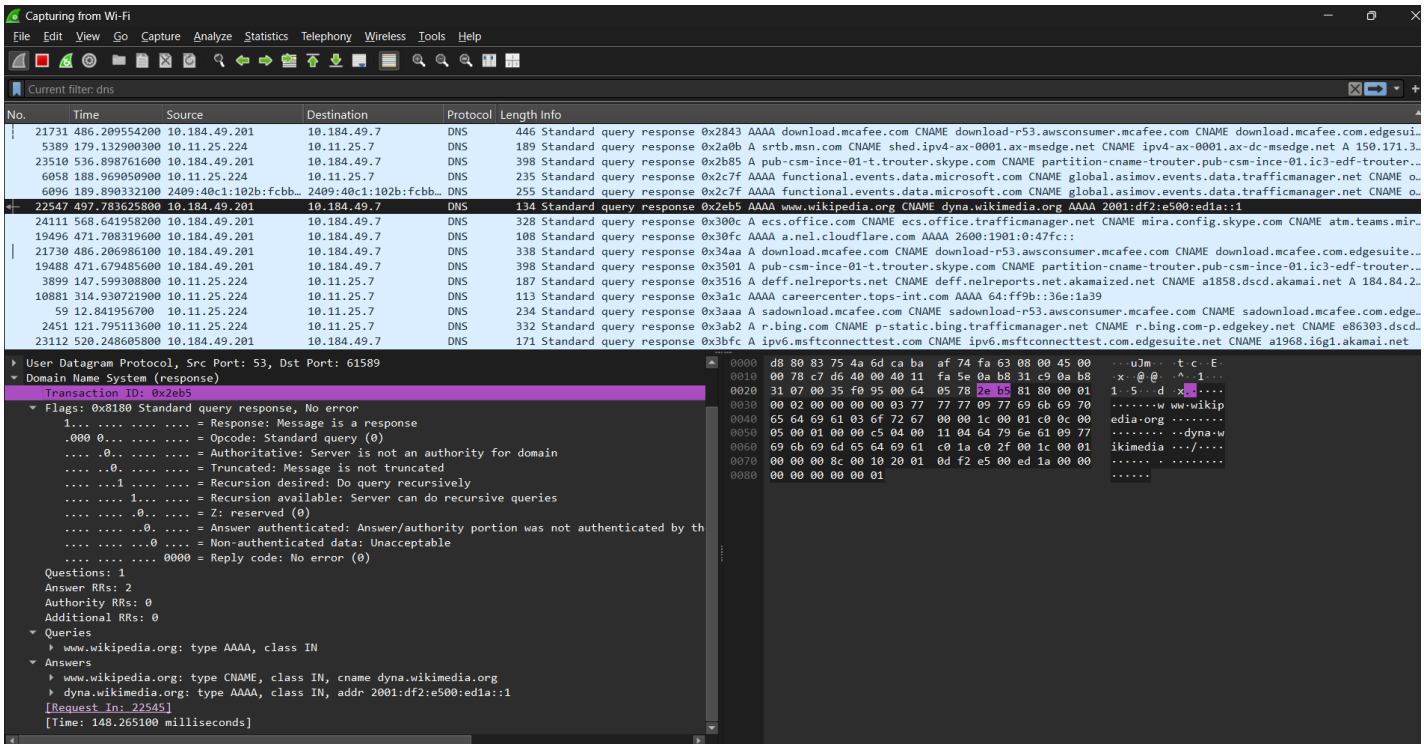
2. Select the active network interface, such as Wi-Fi or Ethernet.
3. Start capturing network traffic.
4. Enter the following display filter in Wireshark:
   dns
5. Open a web browser and visit a website that has not been accessed recently, such as **www.wikipedia.org.**
6. Allow the website to load completely.
7. Return to Wireshark and examine the packets displayed in the packet list.
8. Look for packets containing **Standard query** and **Standard query response** in the Info column.
9. Select the packets and examine their DNS details in the packet details pane.
10. Stop the capture after observing the DNS traffic.

- **Observation: Sequence of DNS Queries and Responses**
- When a new website is accessed, the following sequence of DNS communication may be observed:

- **Step 1: DNS Query**
- The browser or operating system checks whether the IP address of the requested domain is already available in its DNS cache.
- If the required information is not cached, a DNS query is sent to a DNS resolver to request the domain's IP address.
- **Wireshark Info Column:** Standard query



- **Step 2: DNS Response**
- The DNS resolver processes the query and returns the available DNS information, such as an IP address or a CNAME record.
- **Wireshark Info Column:** Standard query response



- **Step 3: Additional DNS Queries (If Required)**
- Sometimes, the DNS response contains a CNAME record pointing to another domain. The resolver may perform additional DNS lookups to obtain the final address.

- **Step 4: Website Connection**
- After receiving the required DNS information, the browser can use the resolved IP address to establish communication with the website's server.

- **DNS Packet Observation Table**
- The following table describes the DNS packets to identify in the Wireshark capture.

Sr. No.	Packet Type	Info Column	Description
1	DNS Query	Standard query	The client requests DNS information for the website's domain name.
2	DNS Response	Standard query response	The DNS resolver returns the requested DNS information.

- **Actual Capture Details:**

Parameter	Observation
Website Visited	www.wikipedia.org
Display Filter	Dns
Query Packet Number	22545
Response Packet Number	22547
Queried Domain Name	www.wikipedia.org
Returned DNS Information	CNAME: dyna.wikimedia.org; IPv6 Address: 2001:df2:e500:ed1a::1

- **Explanation**
- DNS communication generally follows a query-and-response process. First, the client sends a DNS query requesting information about a domain name. The DNS resolver then returns a response containing the available DNS records.
- This process allows the browser to obtain the information required to locate the website's server. DNS caching can reduce repeated queries, so a fresh query may not appear every time a website is opened.

- **Result**
- DNS traffic was captured and examined using Wireshark. DNS packets containing Standard query and Standard query response information were studied to understand the process of domain name resolution.

❖ **5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).**

➢ **Aim**

- To understand the role of DHCP (Dynamic Host Configuration Protocol) through a real-world scenario of connecting a mobile phone to café Wi-Fi and using Instagram.

➢ **Introduction**

➢ **DHCP (Dynamic Host Configuration Protocol)** is a network protocol used to automatically assign IP addresses and other network configuration details to devices connected to a network.

➢ It provides important information such as:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

➢ DHCP reduces the need for manual IP configuration and makes it easier for devices to connect to a network.

➢ **Real-World Scenario: Using Instagram on Café Wi-Fi**

➢ Imagine that a person visits a café and wants to use Instagram on their mobile phone. The café provides a wireless Wi-Fi network.

➢ The following steps explain how DHCP works in this situation:

1. **Connect to Wi-Fi:** The user selects the café's Wi-Fi network on their mobile phone.
2. **DHCP Discover:** The phone sends a DHCP Discover message to find an available DHCP server on the network.
3. **DHCP Offer:** The DHCP server responds with an offer containing an available IP address and other network configuration details.
4. **DHCP Request:** The phone sends a DHCP Request message to request the offered IP address.
5. **DHCP Acknowledgment:** The DHCP server sends a DHCP ACK message to confirm the assignment.
6. **Internet Access:** After receiving the network configuration, the phone can communicate through the café's router and access Internet services such as Instagram.

- **DHCP Communication Process**
- DHCP generally follows the DORA process:

Step	DHCP Message	Purpose
1	Discover	Client searches for a DHCP server.
2	Offer	Server offers an available IP address.
3	Request	Client requests the offered address.

4	Acknowledgment (ACK)	Server confirms the IP address assignment.

- **Importance of DHCP in This Scenario**
  - Automatically assigns an IP address to the mobile phone.
  - Reduces manual network configuration.
  - Helps multiple customers connect to the café's Wi-Fi.
  - Provides gateway and DNS information required for network communication.
  - Makes connecting to public and private networks easier.

- **Result**
- The use of DHCP was explained through the example of connecting a mobile phone to café Wi-Fi and accessing Instagram. The four steps of DHCP communication—Discover, Offer, Request, and Acknowledgment—were studied.